

Do Electrothermal Models Transport Across PMSMs?

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This supplement reports the complete frozen evidence behind the main manuscript. Temperatures are in degrees Celsius unless stated otherwise. One complete operating profile is the statistical unit; seconds within a profile are dependent trajectory points. “External” means the second recorded IPMSM, not a fleet sample. No table below removes predeclared-unsupported profiles from a primary endpoint.

The source dataset contains 1,330,816 rows, 69 profiles, one 52 kW PMSM, 2 Hz native acquisition, and 184.836 audited hours. The external dataset contains 97,725 rows, 16 profiles, one IPMSM, and a published one-second update step. The raw external rows imply 27.146 h; the source repository states 23.8 h. The discrepancy was frozen as a metadata limitation, and timing calculations use row order and the code-defined step.

The grouped source split is immutable:

- train (44): 3, 4, 5, 8, 9, 15, 16, 17, 18, 19, 21, 24, 27, 29, 30, 36, 41, 44, 46, 47, 48, 49, 50, 51, 52, 57, 58, 60, 61, 62, 64, 66, 67, 68, 70, 72, 73, 74, 75, 76, 78, 79, 80, 81;
- validation (11): 10, 13, 23, 31, 32, 42, 53, 54, 55, 63, 69;
- locked test (14): 2, 6, 7, 11, 12, 14, 20, 26, 43, 45, 56, 59, 65, 71.

All 16 external profiles were locked from selection. External aggregate columns `active_wind_est`, `stator_est`, and `rotor_est` are outputs of the published target-specific model. They were never loaded into the primary feature, target, tuning, support, or evaluation pipeline. Raw `id_*.csv` files were authoritative. Their shared 22 columns agreed exactly with the same raw columns in the aggregate file.

The harmonized state was winding, stator core, and rotor. Source `stator_winding` mapped to the mean of external `activewind_1` and `activewind_2`; the mean of source `stator_tooth` and `stator_yoke` mapped to the mean of external `slotbottom` and `outer_yoke`; source `pm` mapped to external `rotor`. The last mapping is a semantic proxy and not an identical sensor location.

Source samples were reduced to 1 Hz by averaging paired exogenous/boundary values and retaining the thermal endpoint. Every primary profile exposed seconds $[0, 300)$ for commissioning. The state at second 299 anchored a recursive hidden-label rollout over $[300, 1500)$. Source profiles also used $[300, 3600)$ for the long-horizon endpoint. Frozen models could use the boundary state but could not fit to prefix temperatures. Target-only and source-prior models could use only prefix transitions.

For thermal node j , the constrained model was

$$\Delta T(j, t) = \sum_{k \neq j} a(j, k) [T(k, t-1) - T(j, t-1)] + b(j, c) [T_c(t) - T(j, t-1)] + b(j, a) [T_a(t) - T(j, t-1)] + \sum_{m=1 \dots 5} q(j, m) \phi_m(t).$$

Every coefficient was nonnegative. Loss proxies were $|i|^2$, $|i|^2|\omega|$, $|u||i|$, $|\tau\omega|$, and $|\omega|^2$. The raw model retained source absolute scale through source-only numerical denominators. The normalized model used only first-prefix exogenous magnitudes and source-learned positive floors. These coefficients are effective grey-box terms, not uniquely identified physical resistances or capacitances.

Profile-macro RMSE first averaged the three node RMSEs within each profile and then equally averaged profiles. Paired intervals resampled complete profiles 10,000 times. The common numerical guard was $[-50, 250]$ °C; a clipped prediction remained in the error calculation and every clip was counted.

The first five rows tune Ridge alpha; the remaining rows tune the quadratic source-prior penalty. Only 11 source-validation profiles entered this table. Source test and all external errors were unknown at selection.

Family	Candidate	Validation RMSE	Clips	Selected
Ridge Alpha	0.01	14.087	0	no
Ridge Alpha	0.1	14.132	0	no
Ridge Alpha	1.0	14.238	0	no
Ridge Alpha	10.0	13.826	0	no
Ridge Alpha	100.0	10.543	0	yes
Source Prior Penalty	0.1	13.427	0	yes
Source Prior Penalty	1.0	25.158	48	no
Source Prior Penalty	10.0	18.485	111	no
Source Prior Penalty	100.0	15.621	0	no
Source Prior Penalty	1000.0	14.439	0	no

The frozen values were Ridge alpha 100 and source-prior penalty 0.1. The selected model bundle, coefficient table, validation table, configuration, and selection metadata were written before the locked outcomes. A later fast tree-traversal implementation changed only inference speed for the already fitted histogram residual comparator; it matched public scikit-learn predictions to absolute tolerance 10^{-14} and preceded any test outcome.

The table contains every locked method. Mean, median, and worst values are complete-profile macro RMSE. The published LPTN row is external context only because it uses target-specific development information unavailable to the target-blind methods.

Method	Source mean	Source median	Source worst	Source clips	External mean	External median	External worst	External clips
Initial-state persistence	14.303	12.416	26.249	0	6.385	6.664	13.124	0
Boundary-shift persistence	13.555	12.141	29.214	0	6.166	6.274	12.609	0
Source Ridge ARX	9.570	7.867	17.013	0	16.760	8.072	86.487	306
Raw source thermal network	1.867	1.786	4.068	0	24.493	27.280	44.321	0
Normalized source thermal network	8.918	8.507	22.469	0	21.984	9.786	122.962	683
Target-only prefix network	6.023	5.086	17.859	0	6.018	5.205	23.499	0
Source-prior prefix network	13.508	6.270	58.534	45	13.610	4.291	151.018	999
Source thermal + residual	9.153	9.202	20.288	0	26.128	17.719	120.950	684
Published target-specific LPTN (context)	-	-	-	-	3.468	3.067	8.393	0

The raw source network's 1.867 °C source error and 24.493 °C external error give a 13.1167-fold transport degradation. The source-prior method's external median (4.291 °C) is much smaller than its mean (13.610 °C) because profile 14 is catastrophic. Therefore medians alone would conceal a deployment failure.

S4.1 Per-node results for the principal thermal models

Dataset	Method	Node	Mean RMSE	Mean MAE	Mean max error	Clips
External	Normalized source thermal network	Rotor	12.236	9.939	22.320	683
External	Normalized source thermal network	Stator Core	17.796	15.192	26.963	683
External	Normalized source thermal network	Winding	35.920	31.720	51.526	683
External	Raw source thermal network	Rotor	9.887	8.557	16.200	0
External	Raw source thermal network	Stator Core	20.245	17.889	31.106	0
External	Raw source thermal network	Winding	43.346	38.815	69.330	0
External	Source-prior prefix network	Rotor	12.019	10.039	20.851	999
External	Source-prior prefix network	Stator Core	12.548	10.666	19.396	999
External	Source-prior prefix network	Winding	16.265	14.778	21.618	999
External	Target-only prefix network	Rotor	8.227	7.078	13.986	0
External	Target-only prefix network	Stator Core	5.697	4.548	11.921	0
External	Target-only prefix network	Winding	4.128	3.328	8.500	0
Source	Normalized source thermal network	Rotor	6.331	5.417	10.614	0
Source	Normalized source thermal network	Stator Core	6.644	5.767	10.886	0
Source	Normalized source thermal network	Winding	13.779	12.124	22.450	0
Source	Raw source thermal network	Rotor	1.889	1.587	3.665	0
Source	Raw source thermal network	Stator Core	1.516	1.306	2.799	0
Source	Raw source thermal network	Winding	2.197	1.886	4.205	0
Source	Source-prior prefix network	Rotor	11.035	9.402	19.474	45
Source	Source-prior prefix network	Stator Core	9.988	8.716	17.408	45
Source	Source-prior prefix network	Winding	19.500	16.967	34.656	45
Source	Target-only prefix network	Rotor	5.750	4.845	10.355	0
Source	Target-only prefix network	Stator Core	4.847	4.109	8.075	0
Source	Target-only prefix network	Winding	7.472	6.303	13.205	0

Source permanent-magnet and external rotor temperatures are displayed under the common “Rotor” label only for the frozen semantic bridge. The large raw-model external winding error should not be interpreted as a controlled estimate of sensor-location effects.

S4.2 Complete source-prior paired comparisons

Positive improvement means the candidate source-prior method has lower RMSE than the named baseline. Intervals and Holm-adjusted values are profile-level descriptions; they do not provide machine-population inference.

Dataset	Horizon	Baseline	Baseline RMSE	Prior RMSE	Improvement	95% interval	Holm p
External	matched 20min	Boundary-shift persistence	6.166	13.610	-7.444	[-26.762, 2.853]	0.1160
External	matched 20min	Initial-state persistence	6.385	13.610	-7.225	[-26.629, 3.138]	0.1160
External	matched 20min	Normalized source thermal network	21.984	13.610	8.373	[0.130, 17.652]	0.1869
External	matched 20min	Raw source thermal network	24.493	13.610	10.882	[-12.251, 25.296]	0.0361
External	matched 20min	Source Ridge ARX	16.760	13.610	3.150	[-7.449, 11.039]	0.0549
External	matched 20min	Source thermal + residual	26.128	13.610	12.517	[3.736, 21.215]	0.0377
External	matched 20min	Target-only prefix network	6.018	13.610	-7.593	[-23.847, 0.948]	0.6322
Source	long 55min	Boundary-shift persistence	16.576	28.654	-12.078	[-34.385, 4.692]	1.0000
Source	long 55min	Initial-state persistence	18.722	28.654	-9.933	[-31.449, 6.148]	1.0000
Source	long 55min	Normalized source thermal network	13.252	28.654	-15.403	[-38.083, 2.603]	1.0000
Source	long 55min	Raw source thermal network	2.376	28.654	-26.278	[-47.258, -10.067]	0.0043

Dataset	Horizon	Baseline	Baseline RMSE	Prior RMSE	Improvement	95% interval	Holm p
Source	long 55min	Source Ridge ARX	13.226	28.654	-15.429	[-36.942, 1.572]	1.0000
Source	long 55min	Source thermal + residual	12.880	28.654	-15.774	[-37.732, 1.009]	1.0000
Source	long 55min	Target-only prefix network	10.943	28.654	-17.712	[-38.955, -1.010]	0.7134
Source	matched 20min	Boundary-shift persistence	13.555	13.508	0.047	[-10.595, 8.507]	0.9185
Source	matched 20min	Initial-state persistence	14.303	13.508	0.795	[-9.397, 8.500]	0.9185
Source	matched 20min	Normalized source thermal network	8.918	13.508	-4.590	[-15.642, 4.079]	1.0000
Source	matched 20min	Raw source thermal network	1.867	13.508	-11.640	[-21.536, -3.902]	0.0120
Source	matched 20min	Source Ridge ARX	9.570	13.508	-3.938	[-12.880, 3.364]	1.0000
Source	matched 20min	Source thermal + residual	9.153	13.508	-4.355	[-15.196, 4.393]	1.0000
Source	matched 20min	Target-only prefix network	6.023	13.508	-7.485	[-17.670, 0.479]	0.9185

Every budget is followed by the same 600-s hidden-label rollout. This prevents later budgets from receiving an easier endpoint merely because their remaining profile segment is shorter. The five-minute values here use a 10-minute rollout and therefore differ from the primary five-minute, 20-minute endpoint.

Dataset	Prefix (min)	Method	Mean RMSE	Median RMSE	Worst RMSE	Clips
External	1	Normalized source thermal network	44.218	5.087	174.082	2294
External	1	Source-prior prefix network	23.931	9.601	157.427	1213
External	1	Target-only prefix network	9.317	5.814	48.755	153
External	5	Normalized source thermal network	12.454	5.513	82.882	83
External	5	Source-prior prefix network	8.090	2.611	89.429	129
External	5	Target-only prefix network	3.408	3.228	7.853	0
External	15	Normalized source thermal network	4.897	4.723	18.642	0
External	15	Source-prior prefix network	2.566	1.831	10.370	0
External	15	Target-only prefix network	2.065	1.819	5.349	0
Source	1	Normalized source thermal network	11.502	7.484	45.082	0
Source	1	Source-prior prefix network	20.649	16.269	66.687	202
Source	1	Target-only prefix network	22.395	16.775	70.917	34
Source	5	Normalized source thermal network	6.019	4.589	17.366	0
Source	5	Source-prior prefix network	7.463	3.127	33.096	0
Source	5	Target-only prefix network	3.523	3.227	8.911	0
Source	15	Normalized source thermal network	6.550	6.147	14.358	0
Source	15	Source-prior prefix network	2.247	1.878	8.133	0
Source	15	Target-only prefix network	2.035	1.043	12.082	0

The 15-minute target-only result is the lowest average error on both recorded machines and has no guard activation. This does not establish an optimal universal budget; it shows that the source prior was not needed once these prefixes supplied sufficient local transitions.

Each source-validation profile contributed one maximum absolute error per node. With 11 calibration profiles and nominal 90% coverage, rank 11-the maximum calibration score-sets the constant half-width. “Node coverage” means every second for the named node was inside its band. “Joint” means all three nodes and all seconds were covered. External values are stress-test outcomes without a cross-machine guarantee.

Dataset	Method	Node	Half-width	Node coverage	Joint coverage
External	Boundary-shift persistence	Rotor	43.099	16/16	16/16
External	Boundary-shift persistence	Stator Core	41.819	16/16	16/16
External	Boundary-shift persistence	Winding	83.229	16/16	16/16
External	Initial-state persistence	Rotor	44.592	16/16	16/16
External	Initial-state persistence	Stator Core	43.121	16/16	16/16
External	Initial-state persistence	Winding	84.539	16/16	16/16

Dataset	Method	Node	Half-width	Node coverage	Joint coverage
External	Normalized source thermal network	Rotor	26.741	13/16	13/16
External	Normalized source thermal network	Stator Core	46.703	13/16	13/16
External	Normalized source thermal network	Winding	91.637	13/16	13/16
External	Raw source thermal network	Rotor	3.810	2/16	1/16
External	Raw source thermal network	Stator Core	4.317	1/16	1/16
External	Raw source thermal network	Winding	8.058	1/16	1/16
External	Source-prior prefix network	Rotor	72.058	15/16	15/16
External	Source-prior prefix network	Stator Core	87.265	15/16	15/16
External	Source-prior prefix network	Winding	158.883	15/16	15/16
External	Source Ridge ARX	Rotor	20.195	13/16	11/16
External	Source Ridge ARX	Stator Core	20.165	11/16	11/16
External	Source Ridge ARX	Winding	44.405	11/16	11/16
External	Source thermal + residual	Rotor	38.455	13/16	13/16
External	Source thermal + residual	Stator Core	51.647	14/16	13/16
External	Source thermal + residual	Winding	100.600	14/16	13/16
External	Target-only prefix network	Rotor	66.992	16/16	15/16
External	Target-only prefix network	Stator Core	37.807	15/16	15/16
External	Target-only prefix network	Winding	77.821	16/16	15/16
Source	Boundary-shift persistence	Rotor	43.099	13/14	13/14
Source	Boundary-shift persistence	Stator Core	41.819	14/14	13/14
Source	Boundary-shift persistence	Winding	83.229	14/14	13/14
Source	Initial-state persistence	Rotor	44.592	13/14	13/14
Source	Initial-state persistence	Stator Core	43.121	14/14	13/14
Source	Initial-state persistence	Winding	84.539	14/14	13/14
Source	Normalized source thermal network	Rotor	26.741	14/14	14/14
Source	Normalized source thermal network	Stator Core	46.703	14/14	14/14
Source	Normalized source thermal network	Winding	91.637	14/14	14/14
Source	Raw source thermal network	Rotor	3.810	10/14	9/14
Source	Raw source thermal network	Stator Core	4.317	13/14	9/14
Source	Raw source thermal network	Winding	8.058	13/14	9/14
Source	Source-prior prefix network	Rotor	72.058	13/14	13/14
Source	Source-prior prefix network	Stator Core	87.265	14/14	13/14
Source	Source-prior prefix network	Winding	158.883	14/14	13/14
Source	Source Ridge ARX	Rotor	20.195	13/14	11/14
Source	Source Ridge ARX	Stator Core	20.165	12/14	11/14
Source	Source Ridge ARX	Winding	44.405	12/14	11/14
Source	Source thermal + residual	Rotor	38.455	14/14	14/14
Source	Source thermal + residual	Stator Core	51.647	14/14	14/14
Source	Source thermal + residual	Winding	100.600	14/14	14/14
Source	Target-only prefix network	Rotor	66.992	14/14	14/14
Source	Target-only prefix network	Stator Core	37.807	14/14	14/14
Source	Target-only prefix network	Winding	77.821	14/14	14/14

The source-prior half-widths are 72.06 °C (rotor), 87.26 °C (stator core), and 158.88 °C (winding). Their 106.07 °C mean makes the 15/16 external joint coverage operationally weak. In contrast, the raw network's mean half-width is 5.40 °C but covers only 1/16 external profiles jointly. Both coverage and sharpness are necessary.

Support used median and 95th-percentile summaries of first-five-minute exogenous variables, source-train robust scaling, and nearest-source-profile distance. The threshold 7.8263 was the 95th percentile of source leave-one-profile-out nearest-neighbour distances. It did not use target temperature labels.

S7.1 External profile decisions

External profile	Distance	Threshold	Decision	Nearest source profile
0	7.099	7.826	supported	18
1	5.544	7.826	supported	30
2	5.676	7.826	supported	30
3	14.581	7.826	rejected	5
4	16.939	7.826	rejected	80
5	15.943	7.826	rejected	16
6	14.330	7.826	rejected	8
7	6.853	7.826	supported	3
8	7.670	7.826	supported	5
9	11.619	7.826	rejected	44
10	8.016	7.826	rejected	19
11	11.577	7.826	rejected	72
12	9.733	7.826	rejected	70
13	6.709	7.826	supported	57
14	10.158	7.826	rejected	3
15	11.794	7.826	rejected	68

Six profiles were accepted and ten rejected. This 62.5% abstention rate is reported beside the accepted-subset accuracy and prevents the subset from replacing the all-profile result.

S7.2 Support-stratified performance

Subset	Method	Profiles	Mean RMSE	Median RMSE	Worst RMSE	Clips
accepted	Boundary-shift persistence	6	5.144	5.367	10.283	0
accepted	Published target-specific LPTN	6	2.099	2.153	3.895	0
accepted	Normalized source thermal network	6	26.452	19.653	52.323	0
accepted	Raw source thermal network	6	19.420	19.488	31.509	0
accepted	Source-prior prefix network	6	3.193	4.218	4.676	0
accepted	Target-only prefix network	6	4.565	4.588	7.374	0
rejected	Boundary-shift persistence	10	6.780	6.274	12.609	0
rejected	Published target-specific LPTN	10	4.289	4.377	8.393	0
rejected	Normalized source thermal network	10	19.303	6.143	122.962	683
rejected	Raw source thermal network	10	27.537	31.208	44.321	0
rejected	Source-prior prefix network	10	19.861	4.807	151.018	999
rejected	Target-only prefix network	10	6.889	5.205	23.499	0

On accepted profiles, source-prior calibration has 3.193 °C RMSE versus 4.565 °C for target-only fitting. On rejected profiles, the corresponding errors are 19.861 and 6.889 °C. All 999 external source-prior clips occur in the rejected subset. This supports further prospective study of gated priors, but one external machine and six accepted profiles cannot certify a deployment rule.

The following are all primary matched-horizon profile/method combinations with at least one clip. No non-finite prediction occurred. Macro RMSE and maximum absolute error are averages across the three nodes for the profile; clip count is accumulated over node-seconds.

Dataset	Profile	Method	Macro RMSE	Macro max error	Clips
Source	45	Source-prior prefix network	48.071	92.819	45

Dataset	Profile	Method	Macro RMSE	Macro max error	Clips
External	14	Source Ridge ARX	86.487	146.268	306
External	14	Normalized source thermal network	122.962	167.731	683
External	14	Source-prior prefix network	151.018	214.101	999
External	14	Source thermal + residual	120.950	163.123	684

Clipping prevents unbounded values from corrupting files; it is not a stability remedy. The guarded errors remain in all rankings. The unedited external profile-14 source-prior trajectory reaches the 250 °C ceiling in winding and stator-core states and exceeds 200 °C in rotor prediction while the recorded states remain near ordinary temperatures.

Artifact	SHA-256 or commit
Protocol configuration	5e98e959b557b69f2c0d2c0f2b035290751ce135cb3a225f8d6b79a8e5c1bc63
Protocol document	ad0834d5e5693e9bc210932a9d41bafead580b7d6ddc40a6f2cde5e4ef336620
Selection freeze	1fa8dc73b38ec597e82862c862062928d33cb0a8272f71788f7df78d854f3bd0
Source raw CSV	78f3d150f0f2ad9c5dc7ff24dd12c00d386ad530f48c1589ad24fcd88867d3ad
External repository commit	98e4566b5fb7c70499996fda18dd73179ec16509
Result: aggregate_summary.csv	03b41560feda8b35a50d8b6ff0411e8ee04a49c7ffdf843a546ec06e151820e
Result: budget_sensitivity_per_profile.csv	ec2f5c8c7d4c9fe890c5cee82c78c439dcac9c4f21d280eed8107b2f799b3bfff
Result: budget_sensitivity_summary.csv	eb453b4e244b4f3ea5607b27c3c0caecadda0d74f5f837f9896645aa96f568
Result: paired_method_comparisons.csv	45ca6e2b86c639f231805968338c49b730b35405bd70dc00cdd9e83a2d78ec72
Result: per_profile_errors.csv	d79ef4c062a179515cbca35a1d916d25e6eac784168a5b31c7f28cd4a5577cd7
Result: predeclared_gates.json	2619c9c8f4bed1914ceeb170f523ac0abfd9718daf3e23c33734693fffdfe6e
Result: selection_freeze.json	1fa8dc73b38ec597e82862c862062928d33cb0a8272f71788f7df78d854f3bd0
Result: source_model_bundle.joblib	5821bd198d5b456d0f8724dfd8d472ff01f1c9c10fd292244ab8a9b2737607
Result: source_model_coefficients.csv	ec0153ca4c9e591ff04644a7891f37960f0ca25c750676313606e4ae6187d52
Result: support_reference.csv	2bd3115a82ed0ece09a85a8ab555e453b00f9e6f83d01685763f4a0e7015377c
Result: support_scores.csv	b3eba9a9045d82aaa689fa2e2f8b6b1b5d36e64c3b9de91b8feeb345ad75faf
Result: trajectory_band_joint_summary.csv	01a843241c9b127b6edbdc651e32d00e803fe04914190631300d4560fae2dfc
Result: trajectory_band_profile_coverage.csv	f948de43ca0789fa679f55002ff191a698e46df2e07e72bb0416412861008c74
Result: trajectory_band_summary.csv	b9b89cf34d4ddc333959ba039d3fac5739d1e09e940fc870596b85d71f4e0691
Result: trajectory_bands.csv	01b3d83a1924d3f76014586375a8966ed116349d16c7d851c2e1c20b7d8108aa
Result: trajectory_predictions.csv.gz	eda4c458b9c4748952c32a7c5d40ac98ac5f33eb58f9d1356de493310d10f76
Result: validation_tuning.csv	7df5a8361230133c8ef9cbfea582be4a63b527ba6558f25963c82cddcd0ecbe6

The audit records Python 3.12.13 (main, Aug 7 2026, 02:26:41) [MSC v.1944 64 bit (AMD64)], NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.0, and scikit-learn 1.9.0. The evidence validator recomputes the raw-data, protocol, configuration, and selection hashes; re-reads the result tables; checks every headline result, gate, support denominator, band width, guard count, figure pair, and bibliography key; and writes a JSON report.

1. Two physical machines do not establish fleet transport. Profile bootstrap intervals describe condition variation for those machines only. 2. The external shift is compound, so the failure cannot be causally assigned to power, cooling, topology, sensors, controller scaling, or excitation separately. 3. Rotor and permanent-magnet sensors are semantic proxies rather than equivalent labels. 4. Target commissioning assumes temporary access to all three temperatures. It is an information benchmark for instrumented commissioning, not a sensor-free production claim. 5. The nonnegative model does not guarantee global closed-loop stability. Numerical guards expose rather than solve this limitation. 6. External support and interval outcomes are descriptive; neither restores formal cross-machine conformance validity. 7. The target-specific published LPTN is contextual and must never be called a target-blind baseline. 8. Post-reveal support-stratified comparisons generate hypotheses only. They do not alter the frozen all-profile endpoints or gates.